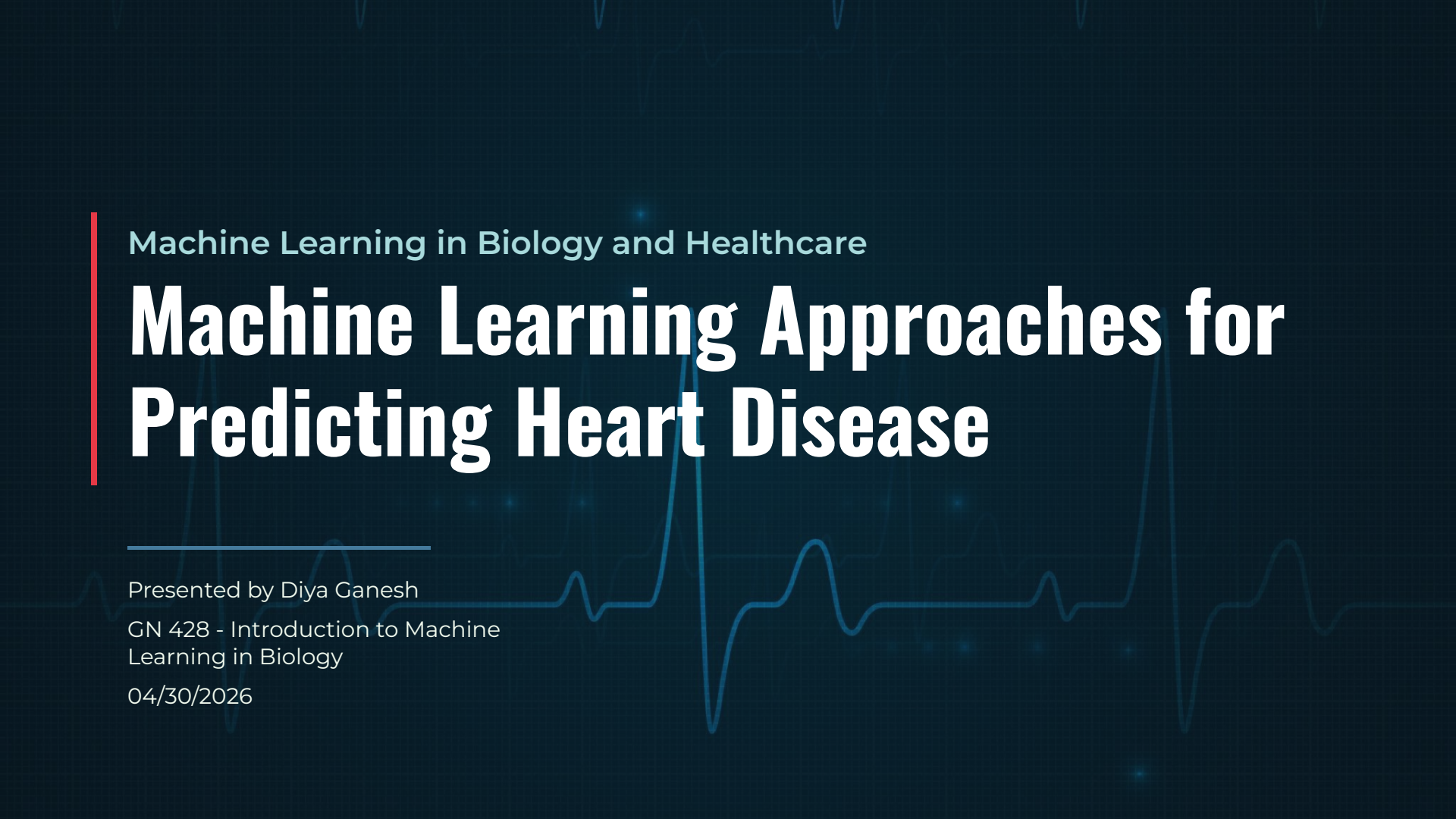




Machine Learning in Biology and Healthcare

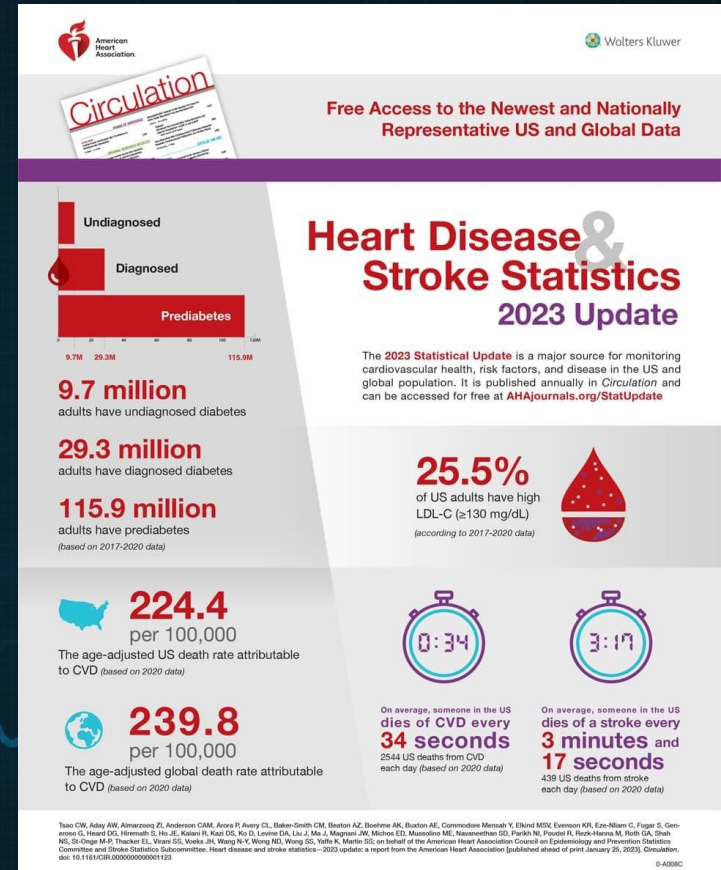
Machine Learning Approaches for Predicting Heart Disease

Presented by Diya Ganesh

GN 428 - Introduction to Machine
Learning in Biology

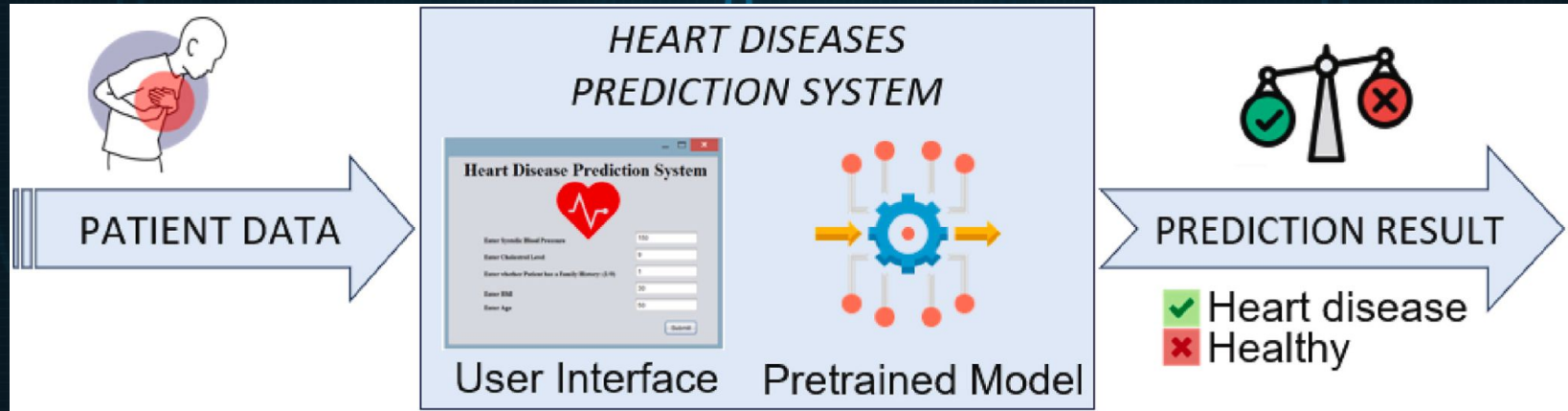
04/30/2026

- Heart disease is a leading cause of death worldwide
- Early detection improves patient outcomes
- Machine learning can analyze clinical patterns
- Goal: predict presence or absence of heart disease



Hypothesis

- Clinical and physiological features are associated with heart disease and that machine learning models can predict disease presence



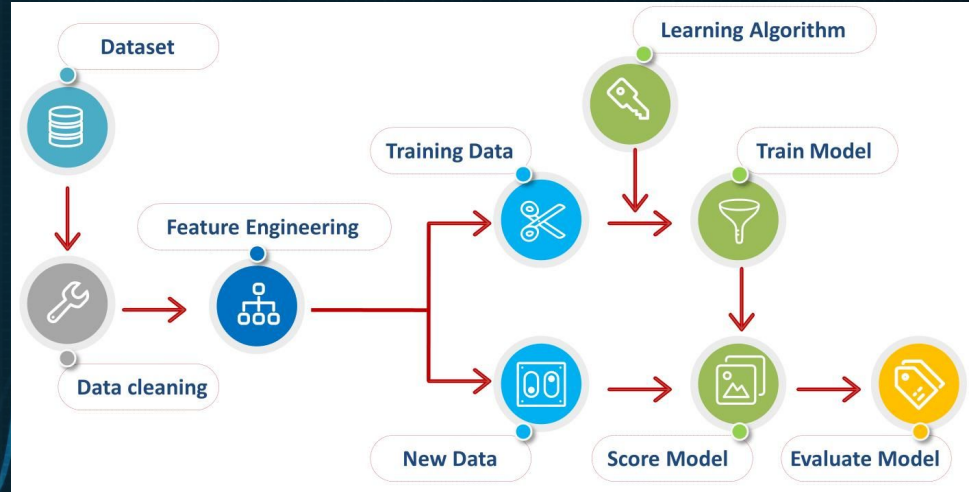
Dataset

- Heart disease dataset
- Outcome:
 - 0 = No heart disease
 - 1 = Heart disease
- Features include:
 - Chest pain type
 - ECG results
 - Exercise-induced angina



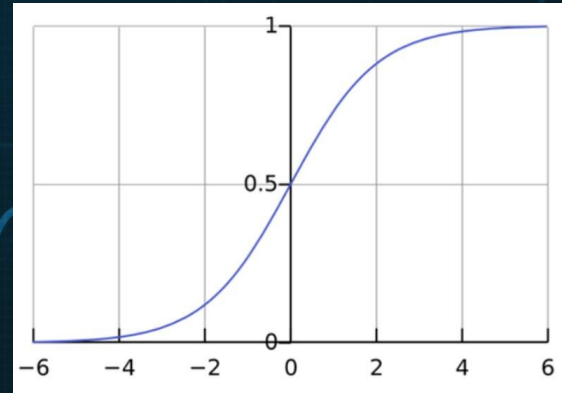
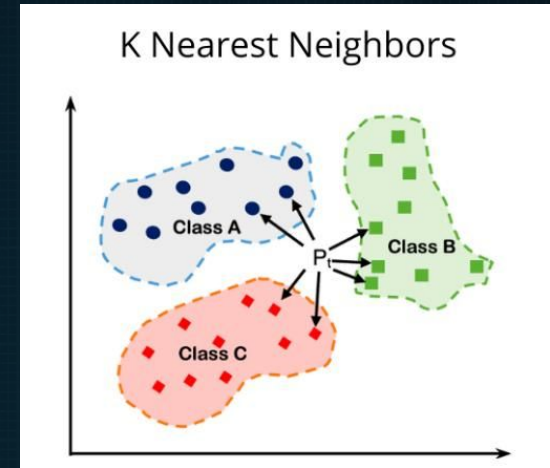
Data Processing

- Converted categorical variables to factors
- Checked for missing values
- Removed duplicate observations
- Split data:
 - 70% training
 - 30% testing

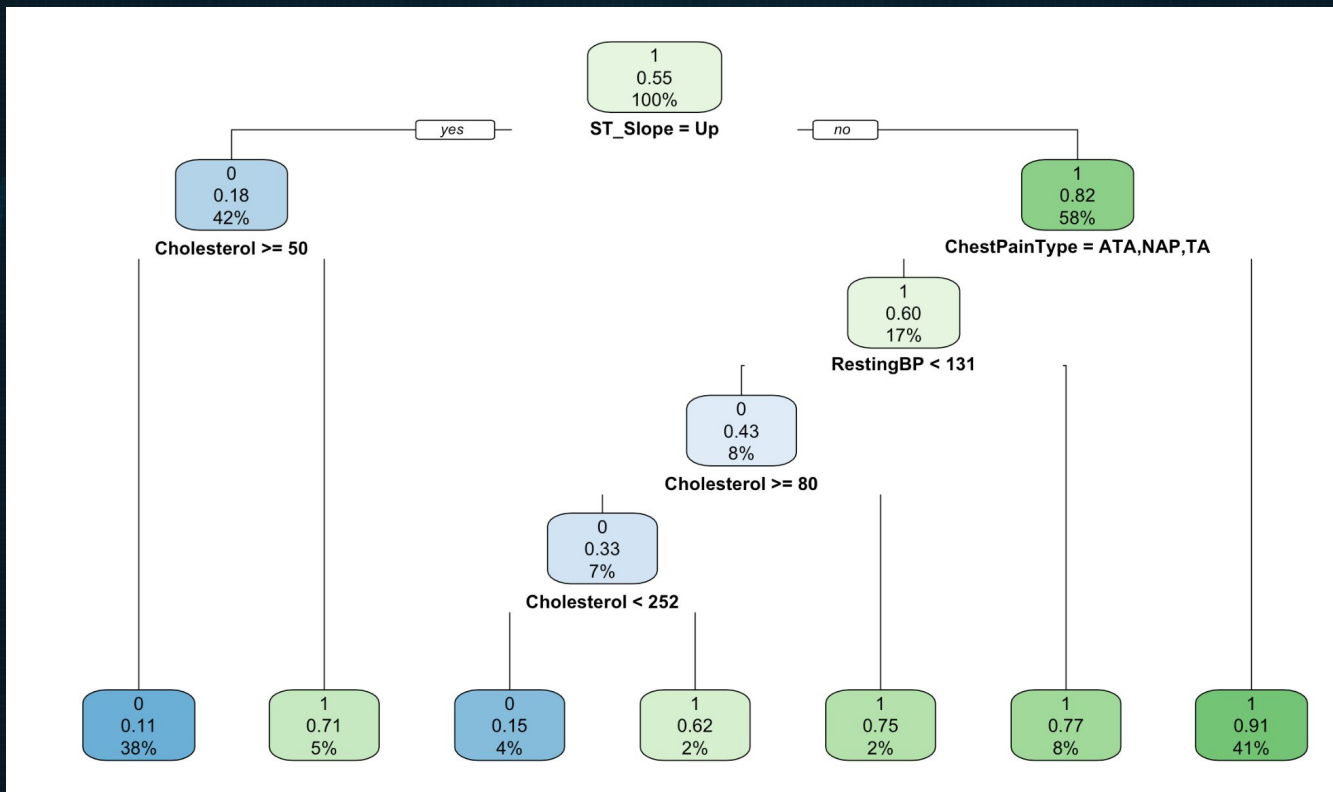


Models Used

- Decision Tree
- Logistic Regression
- K-Nearest Neighbors (KNN)



Decision Tree Model - Figure



Results

Model	Accuracy	Sensitivity	Specificity
Decision Tree	82.97%	83.87%	81.82%
Logistic Regression	87.68%	89.03%	85.95%
KNN	87.68%	90.97%	83.47%

Discussion

- Logistic Regression & KNN had highest accuracy
- KNN had highest sensitivity → best at detecting heart disease
- Sensitivity is important to reduce false negatives
- Decision tree helped identify key features



100

- Future goal: improve sensitivity



References

- fedesoriano. (September 2021). Heart Failure Prediction Dataset. Retrieved [04/25/2026] from <https://www.kaggle.com/fedesoriano/heart-failure-prediction>.